FIELD RECORDING WORKFLOW

Operator’s manual

A practical guide to opening a recording, making careful reversible edits, saving project information, and exporting restored audio.

LOCAL PROCESSING · ORIGINAL AUDIO PRESERVED · WAV OUTPUT

Quick start

01

Open audio
Choose *Import WAV, FLAC or MP4*, or drop a supported file onto the app.

02

Listen and adjust
Play the source, make small edits, and switch between Original and Edited while listening.

03

Save, then export
Save project information in this browser. Export creates a new WAV and a restoration report.

Workspace tour

Field Restore keeps the main controls in a left sidebar, with the recording view and playback controls in the work area.

Area	What it is for
Edit chain	Review, enable or bypass, select, and reorder the processing modules.
Analyze	Compare signal measurements for the original and edited recording, including the current selection.
Project	Edit project title, location, descriptive words, track description, and field notes; access editable project backup controls.
Session controls	Import or relink audio, save the project, load the demo, and open the export dialog.
Top bar	Check current recording and status, undo or redo edits, open Help and settings, and export audio.
Transport	Play, pause, seek, and monitor the current recording. Use the timeline to make a time selection.

Audio operations are performed in the browser on this device; audio is not uploaded for server-side processing. The What3Words location link is external and sends the address you enter to that service when opened.

Import a recording

1. Select **Import WAV, FLAC or MP4** in the sidebar, or drag a file onto the app.
2. Choose a local recording. The original file is read for playback and editing; Field Restore does not overwrite it.
3. Wait for import and analysis to finish. Progress and any errors appear in the work banner.

Supported inputs include uncompressed PCM or 32-bit floating-point WAV, lossless FLAC, and MP4 files with supported audio codecs. MP4 import reads its audio track; it does not edit video. Unsupported codecs or files without a supported audio track will fail with an error.

> **Protect the original**
> Keep the source recording in its original location and make a separate copy before critical work. Edits remain a reversible chain until export.

Inspect and compare

- Use the main plot to inspect the recording over time. Drag across the timeline to select a passage for focused listening and analysis.
- Use the **Original** and **Edited** views to compare the untouched source with the current processing chain.
- Play the full recording and selected passages. Listen for wanted details that might be reduced, not only for quieter background noise.

- Open **Analyze** to compare full-file and selection readings, including true-peak estimates, loudness, and loudness range.

> **Measurements are estimates**
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> These browser-local readings are useful for comparison and review, but are not certified compliance-meter results. Verify critical deliverables in a separate audio application.

Build an edit chain

Edits are ordered, reversible instructions. Use the **+** control beside the chain or open **Modules** to add an edit. Select a module to see its controls. You can bypass a module for comparison, duplicate or remove it, and drag its handle to change its order.

Undo and redo operate on chain changes. Make one modest adjustment at a time, replay the same passage, and compare it with the original before continuing.

Module	Use and caution
Automatic DC correction	Applied first as a required baseline stage; it is not an optional edit.
Gain	Adjusts output level. Watch for clipping and compare at similar listening volume.
Fade In/Out	Softens the beginning or ending. Check that the fade does not hide wanted sounds.
Low Rumble Removal	A fixed high-pass filter: its cutoff applies throughout the recording rather than following automatic wind detection. Use it only for audible rumble; it can also remove wanted bass, thunder, or low calls.
Hum Removal	Targets sustained tones. Adaptive mode can also reduce wanted held sounds, calls, and insects; bypass it to compare.
Wind Removal	Automatically applies approximate low-frequency reduction when wind-like interference is detected. Its response varies over time; wanted bass or low calls may be reduced, some wind may remain, and severe microphone overload cannot be recovered.
Noise Reduction	Use Automatic estimation or select a steady noise-only passage as a sample. Speech, calls, or other wanted audio in a sample may be softened throughout the recording.
Click & Pop Removal	Targets short, isolated clicks and pops. Longer or overlapping sounds may remain; wanted transients can be softened.
Short Sound Repair	Make a localized repair in the selected spectrogram area. Preview the result carefully; it is not a guarantee of perfect reconstruction.

Wind Removal controls

Wind Removal estimates when low-frequency sound resembles wind and applies reduction at those times. Detection is approximate: it can miss wind or react to other low-frequency sounds, and it cannot repair severe microphone overload.

- **Reduction** sets how much low-frequency signal is removed while wind is detected. Higher values remove more.
- **Crossover frequency** sets the upper edge of the affected low-frequency range. Raising it covers a wider range, but can remove more wanted bass and low calls.
- **Smoothing** controls how quickly reduction changes. Higher values make it ramp more slowly as detection starts and stops.

Start with the defaults—72% Reduction, 140 Hz Crossover frequency, and 50% Smoothing—and listen to the same passage. Increase Reduction or Crossover only as needed; if bass or low calls lose weight, lower one or both. Adjust Smoothing if the change sounds abrupt, then compare with the module bypassed before exporting. No setting can guarantee that all wind is removed without affecting wanted sound.

Wind Removal is separate from **Low Rumble Removal**. Wind Removal is automatic and varies with detection; Low Rumble Removal is a fixed high-pass filter that acts throughout the recording at its selected cutoff. Use the fixed filter only when you want consistent low-frequency attenuation, and listen for wanted bass, thunder, or low calls that it may remove.

Processing settings are linked across channels by default to help preserve stereo timing and spatial coherence. Do not judge a repair by its meters alone; listen to the result.

Save and back up a project

Save project

Select **Save project** to store the edit chain, trim, markers, regions, and notes in this browser. This does **not** download a WAV. Saved projects and their source-file permissions are local to this browser and device; reopening may require you to select or relink the original recording.

Back up editable project

In the **Project** workspace, choose **Back up editable project** to download a `.fieldrestore` metadata backup. The backup does not contain audio. Keep it with the original WAV, FLAC, or MP4 file.

To restore, choose **Import project backup** and select the original recording when prompted. Field Restore verifies that the source matches the backup. If it offers **Replace saved project and open**, that replaces the open workspace; back up unsaved work first. Choose **Restore as a copy** if you want to keep the existing saved project.

> **Do not substitute another recording**
> If a project is waiting for its source, relink the original file. If you intend to edit a different recording, open it as a separate recording rather than replacing a pending source unintentionally.

Export audio

1. Review the current chain and listen to the edited result, including important quiet and loud passages.
2. Select **Export 32-bit WAV** and confirm the export.

3. Save both generated files: the restored WAV and its plain-text restoration report.
4. Open the WAV in another player or audio editor to verify playback and channel content.

Export creates 32-bit IEEE-float WAV at the source sample rate and channel count. It does not normalize or apply dither. The original file is not replaced. For MP4 input, only the edited audio is exported as WAV; the video is unchanged.

If a browser download appears stalled or is missing, check the browser's download list and the chosen download location. Do not overwrite the source recording with an export.

Working with MP4

1. Import or drop a supported MP4 file as you would a WAV or FLAC.
2. Field Restore reads the supported audio track locally; video remains unchanged.
3. Edit and compare audio using the same workflow as other recordings.
4. Export produces a WAV containing the edited audio only. It does not replace, modify, or re-encode the MP4.

Some source audio is lossy. Exporting it as WAV does not restore information that was lost in the original encoding. If the codec or track layout is unsupported, use an audio-only file in a supported format and keep the original MP4 unchanged.

Privacy, offline use, and limits

- Audio parsing, playback, analysis, editing, project persistence, and export run locally in the browser. Audio is not uploaded for server-side processing.
- Saved project information is stored in the browser. Clearing site data or using another browser/device can make those local projects unavailable. Keep external backups and original audio.
- Browser memory and local storage limit practical recording size and project persistence. Large recordings can take longer to open, analyze, and export.
- Noise, hum, wind, click/pop, and short-sound repair are conservative approximations. Wanted sounds can be reduced, and some unwanted sound may remain.
- Field Restore does not claim sample-rate conversion, certified loudness compliance, or perfect reconstruction.

> **Windows installer notice**
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> The available Windows installer is unsigned and has not been natively validated for this build. Windows may show an unknown-publisher warning. Do not treat that warning as proof of malware or proof of safety; obtain the installer only from the official Field Restore site and review its published checksum.

Troubleshooting

What you see	What to do
Import fails or the codec is unsupported	Confirm the file opens in another player. Try a supported WAV or FLAC copy. Keep the original unchanged.
A project asks for its source file	Select the original matching WAV, FLAC, or MP4. Do not choose a different recording to satisfy a relink prompt.
A sound becomes dull, thin, or unnatural	Bypass the most recent module, compare Original and Edited at similar volume, then reduce or remove the edit.
A download is not visible	Open the browser’s download history and check its save location and blocked-download notices. Export again only after confirming the first file is not still being written.
Saved project information is missing on another device	Project saves are browser-local. Transfer the `.fieldrestore` backup and original recording manually, then restore with the matching source.
Playback or export is slow	Wait for the progress banner to finish. Large files use more memory and may take longer; avoid reloading the page during an active operation.

Keyboard shortcuts

Shortcut	Action
`Space`	Play or pause
`⌘ Z` / `Ctrl Z`	Undo the last edit-chain change
`⇧ ⌘ Z` / `Ctrl Shift Z`	Redo
`Esc`	Close an open dialog or menu

For a critical restoration, keep the source and exported files, review the restoration report, and verify the WAV in a second audio application.